

Figure 1: Weak polyglot

multiple NoSQL/SQL databases. Supporting multiple data stores is polyglot persistence. A polyglot persistence can have implementation approaches that have weak or strong mapping with data sources.

The figures illustrate weak polyglot and strong polyglot approaches. In the weak polyglot approach, the data access layer directly fetches data from various data sources. However, the strong polyglot has a polyglot mapper to deal with various data sources. A strong polyglot is the recommended solution for scalability and extensibility of the system. There are solutions which support a polyglot mapper. An example is Kundera, an open source framework/library backed by Impetus Labs or iLabs, the R&D division of Impetus Technologies.

Kundera

Kundera is a JPA (Java persistence API) 2.1 compliant polyglot object mapper. It supports various NoSQL databases (Cassandra, MongoDB, HBase, Redis, Oracle NoSQL, Neo4j, Couchdb) and relational databases. It has features like polyglot persistence, JPQL (Java persistence query language) and native query support, schema generation, transaction management, REST based access and many more. Kundera has been developed with the objective of making access across data stores as easy as changing a configuration.

Figure 3 illustrates the solution overview for next generation mobile applications. As stated, an enterprise may have multiple SQL or NoSQL data sources. A polyglot mapper (like Kundera) may provide support for various data sources. This polyglot mapper interacts with an enterprise's CRM (customer relationship management), SCP (supply chain management), ERP (enterprise resource planning) and FSM (field service management) systems. The core enterprise system exposes services or applications, as per the needs, through a secure enterprise data gateway. The exposed services are accessed by mobile applications.

Mobile solutions

Mobile applications have changed the focus from what's on the Web to the applications on mobile devices. So mobile apps are

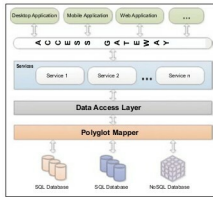


Figure 2: Strong polyglot

no longer an option; they are now an imperative, with measurable ROI (return on investment) for enterprises. With the stress on gaining competitive advantage, ensuring user delight, building an efficient and productive workforce and working towards resource optimisation, there are diverse requirements for enterprise mobility solutions, as listed below.

Data security: Mobility solutions require access to enterprise data. In some cases, the

application saves enterprise data locally on the user's personal device. The security mechanism for 'data on the move' as well as 'data at rest' is one of the crucial concerns for enterprises.

- With data on the move, the concern is with data security during network communication for client-server interaction
- With data at rest, the issue is with data security during storage on mobile devices (required for offline data support from mobile applications)

Enterprise data gateway: Access to enterprise data over generic Wi-Fi or the operator's network is a core requirement for modern mobile solutions. These gateways should have the intelligence to offer enterprise data on only registered devices. However, the data sources may be standard or custom data sources.

Device/platform diversity: Multi-platform supported applications for a BYOD (bring your own device) or CYOD (choose your own device) policy is an unavoidable requirement for enterprises. COPE (corporate owned personally enabled) devices might also be a mode of delivery. Selecting the appropriate development strategy to deal with platform diversity and device diversity is one of the important decisions. It can significantly affect the TCO (total cost of ownership). Applications for in-house distribution, with the admin desk managing user devices in a remote fashion, also add to the TCO.

Manageability (application distribution and data management): Remote data management and applications management are the other expensive dimensions of the enterprise mobility ecosystem.

User experience: Customer or user delight is the way to go with offerings and services. Mobile solutions come with diverse platform-specific guidelines and user interaction models like the Windows Metro guidelines, the iOS Human Interface Guidelines, the Android UX guidelines, etc. Mobile applications can be designed with an app-centric design or adaptive design.

- **App-centric designs:** Mobile applications designed with common guidelines and used across multiple platforms
- **Adaptive designs:** Mobile applications designed with platform-specific guidelines for the desired cross-platform usage